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S.E. (Computer) (Sem. II) EXAMINATION, 2018
OBJECT ORIENTED AND MULTICORE PROGRAMMING
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Neat diagrams must be drawn wherever necessary.

(ii) Figures to the right side indicate full marks.

(iii) Use of Calculator is allowed.

(iv) Assume suitable data, if necessary.

1. (a) Write short notes on : [8]

(i) Need of object-oriented programming

(ii) Dynamic memory allocation.

(b) Explain array of objects with example. [4]

Or

2. (a) Write a C++ program for vector addition using operator overloading. Vector consists of 2 attributes *ax*, *ay* for magnitude and direction (both int). Create 3 vectors v_1 , v_2 , v_3 with v_1 (8, 13) and v_2 (26, 7). After performing $v_3 = v_1 + v_2$; user should be able to print v_3 's *ax* and *ay* values to 34 and 20 resp. [8]

(b) Differentiate between public, private and protected members. [4]

P.T.O.

3. (a) A warehouse management system requires taking user input and displaying items which are present. Use any STL (vector, list, etc) to implement the system. Item consist of 3 attributes (name, code both strings and price in float). Write menu driven C++ program to accept and display items. [8]

- (b) Write a short note on multiple exception handling. [4]

Or

4. (a) Create child processes using `posix_spawn( )` function. Use object oriented approach for process management. Write menu driven C++ program to create n processes (where n is any +ve integer given by user) and display their pid's on console. All n child processes will execute the `ps` utility, which resides in `"/bin/ps"`. [6]

- (b) Write in detail about anatomy/structure of a process. [6]

5. (a) What are the similarities between threads and processes ? [6]

- (b) Explain in detail pthread attribute object. [7]

Or

6. (a) Differentiate between threads and processes. [6]

- (b) Write a detailed note on termination of threads. [7]

7. (a) What is persistence of an object ? Explain persistence with respect to IPC. [4]

(b) Write short notes on IPC mechanism using : [9]

(i) Files

(ii) Shared memory

(iii) Pipes.

Or

8. (a) Explain PRAM model used for synchronization. [4]

(b) Explain the following : [9]

(i) Basic semaphore operations with P() and V().

(ii) Mutex semaphores in POSIX.

(iii) Delegation model for threaded application.